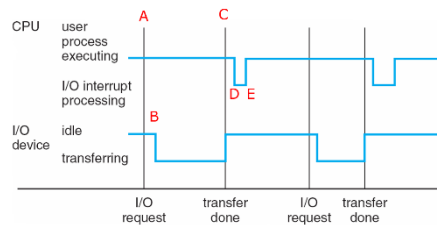


Midterm Exam 2018 (Operating System)

答案請寫在答案卷，題目無須繳回，除非是非題外請附解釋或提供計算過程，2~16 題，每題 2 分滿分:120

1. (5%)選課班別為 ____ (A:上午班, B:下午班)，組別為 ____。
2. **F**(True of False)使用 micro-kernel approach 的 Operating System 也比較不佔 memory space。
3. **F**(True of False) During the disk reading operation, disk controller is responsible(負責) for moving data from controller's buffer to main memory.
4. **T**(True of False) Suppose that *gvar* is a global variable, then *gvar* is independent (獨立) among sibling threads。
5. **T**(True of False) User process may execute user code as well as kernel code 使用者行程，會執行使用者的程式碼，也會執行核心的程式碼。
6. **F**(True of False)The "breakpoints" used in debugger is classified(分類) as "asynchronous" interrupt.
7. **T**(True of False) The execution of interrupt service routine is supposed(應該) to be very quick.
8. **F**(True of False) Compared to API, system call is easier to port(移植) to different platform(平台).
9. **F**(True of False)To reduce(減少) the waiting time, the disk scheduler is in favor(喜好) of I/O-bound process.
10. **F**(True of False)If two threads corresponding(對應) to same process, they share a common stack.
11. **T**(True of False)Windows XP OS has no long-term scheduler.
12. **CD**(複選)Which of the following instructions should be privileged? (a)write to memory (b)issue system call (c)turn off interrupt (d)modify PCB (e)read from memory
13. **A** 以下四種 scheduling algorithm 的評估方式，最適合教學，展示使用的是:
(a)Deterministic modeling, (b)Queueing model, (c)Simulation, (d)Implementation
14. **C** Which system call runs an executable file in the context of an already existing process, replacing the previous executable?



(a)fork(), (b)exit(), (c)exec(), (d)wait()

15. **B** Since which system, keeping CPU busy becomes a major system goal?由哪一個系統開始讓 CPU 越忙越好?
(a)batch (b)multiprogramming (c)time sharing (d)multi-processor
16. **D** 參考右圖，哪一段時間系統執行 interrupt handler?
(a)A-B, (b)B-C, (c)C-D, (d)D-E, (e)E-A

17. (5%) Why is the separation(分開) of mechanism and policy desirable?(希望的)
 Mechanism and policy must be separate to ensure that systems are easy to modify. No two system installations are the same, so each installation may want to tune the operating system to suit its needs. With mechanism and policy separate, the policy may be changed at will while the mechanism stays unchanged. This arrangement provides a more flexible system.
18. (5%) 說明負載平衡(load balancing)和處理器親和性(processor affinity)在利益上為何有衝突。
19. (5%) 說明 Multi-threaded processor 為何能解決 memory stall 帶來的問題
20. (4%) If there are many registers in processor, how can we exploit(利用) it to reduce(降低) the overhead(額外開銷) of context switching?
21. (4%) "To learn OS, one has to study system calls", do you agree or not? Please explain.
22. (4%) Please compare "concurrency"(並行) and "parallelism"(平行)。
23. (4%) One of the benefit in using multithreaded programming is "Resource Sharing". Please explain it by example. Threads share resource of process by nature. Two processes can communicate each other by "shared memory" or "message passing". However, sibling threads communicate each other by data section. It is easier than process communication.
24. (4%) In Solaris, what condition will a time sharing thread upgrade its priority ? what condition demote its priority? 升級與降級的條件 return from sleeping, time quantum expired
25. (10%) The program below uses Pthreads API. What would be the output from **LINE C** and **LINE P 7, 0**

```
int value = 0;
void *run(void *param);
int main(int argc, char *argv[])
{ int pid;
  pthread_t tid;
  pthread_attr_t attr;
  pid = fork();
  if (pid == 0) {
    pthread_attr_init(&attr);
    pthread_create(&tid, &attr, run, NULL);
    pthread_join(tid, NULL);
    printf("CHILD: value = %d", value); /*LINE C*/
  } else if (pid > 0) {
    wait(NULL);
    printf("PARENT: value = %d", value); /*LINE P*/
  }
}
```

```
void run(void *param){
```

```

value = 7;
pthread_exit(0);
}

```

26. (20%) The attributes of five processes are shown in the table below. (a) If 11 unit of CPU time was allocate(分配) to execute Process A, what are the possible states of Process A? (行程 A 的可能狀態有哪些?) (b) Schedule these processes by FCFS, draw the Gantt chart, and calculate the waiting time of each process. (c) same as before but using preemptive SJF, (d) same as before but using Round-Robin with time quantum=4. 畫出 Gantt chart，算出每一行程的等待時間。

Process	Arrival Time	Burst Time
A	0	11
B	2	8
C	5	28
D	9	16
E	13	2

(time, Process)

FCFS: (0,A)-(11,B)-(19,C)-(47,D)-(63,E)-(65,null), end time=(11,19,47,63,65)

SJF: (0,A)-(11,B)-(19,E)-(21-D)-(37,C)-(65,null), end time=(11,19,65,37,21)

SRTF: (0,A)-(2,B)-(10,A)-(13,E)-(15,A)-(21,D)-(37,C)-(65,null), end time=(21,10,65,37,15)

RR(q=4): (0,A)-(4,B)-(8,A)-(12,C)-(16,B)-(20,D)-(24,A)-(27,E)-(29,C)-(33,D)-(37,C)-(41,D)-(45,C)-(49,D)-(53,C)-(57,C)-(61,C)-(65,null), end time=(27,20,65,53,29)

27. (20%) Explain the following terms. Please demonstrate your knowledge in associate with those terms, instead of translations. Examples are very welcome. Clear description helps credit. (a)context switch (b)Amdahl's law (c)soft realtime system (d)trace tape

解釋下列名詞。請展現對於作業系統的理解程度，切勿翻譯了事。盡可能多用例子。字數太少，資訊不足都很可能導致無法正確的評估。因此，請盡量多寫，寫清楚。惜墨如金的，責任自負。